What is Keyword Spotting?

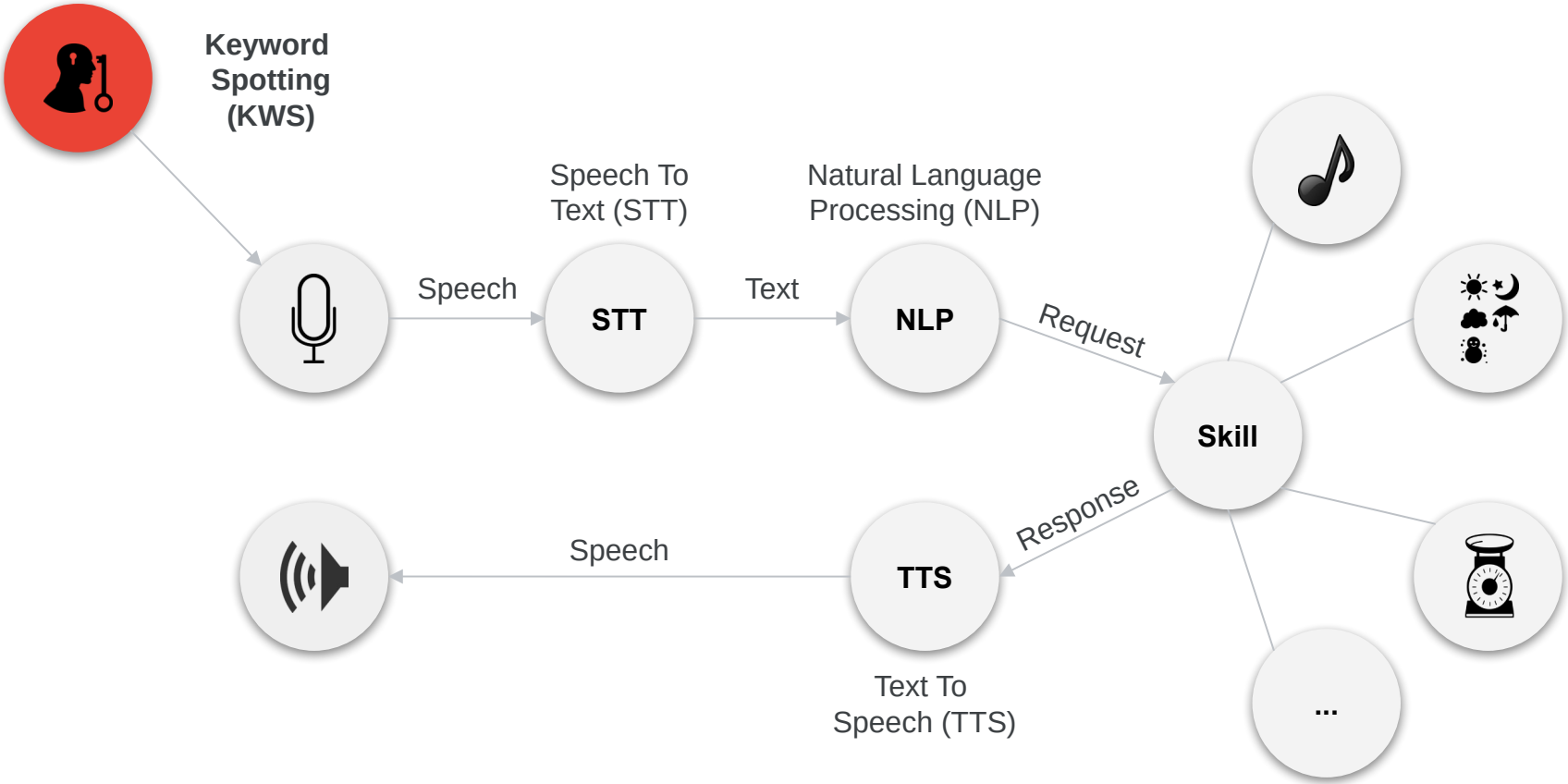
Keyword Spotting v. General Speech Recognition

- **Keyword spotting** is one of the most successful examples of **TinyML**
  - Low-power, continuous, on-device
  - Common Voice SWTS expands keyword spotting to more languages

Keyword Spotting v. General Speech Recognition

- **Keyword spotting** is one of the most successful examples of **TinyML**
  - Low-power, continuous, on-device
  - Common Voice SWTS expands keyword spotting to more languages

- **General ASR** still requires **larger, power-hungry models**
  - But it can run on mobile devices (offline dictation on smartphones)


Keyword Spotting (KWS)
Speech To Text (STT)
Natural Language Processing (NLP)
Speech
STT
Text
NLP
Request
Skill
Response
Speech
TTS
Text To Speech (TTS)
...


amazon

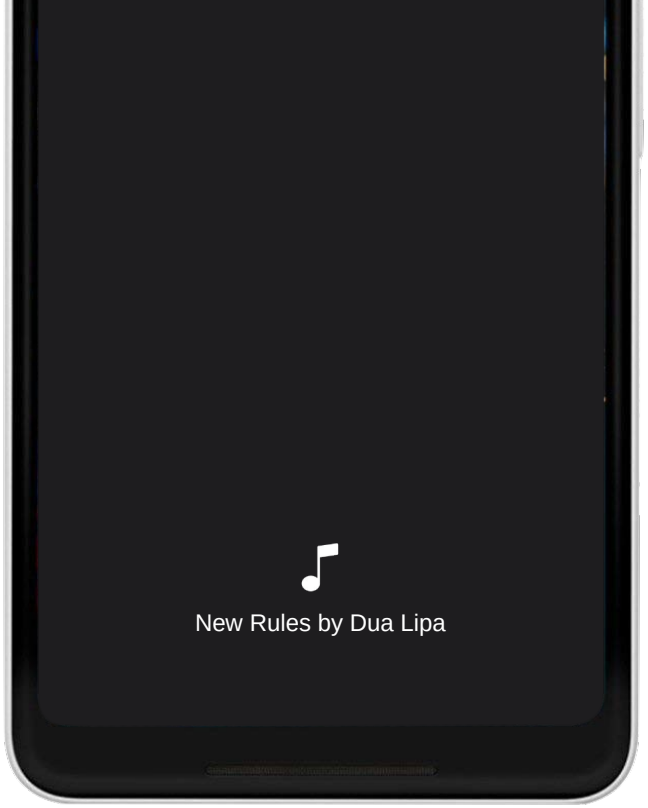




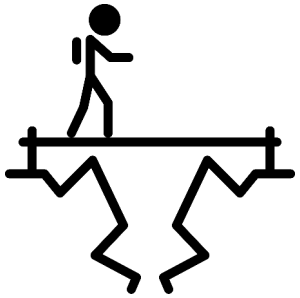


What else can **TinyML** do with audio data?

Now Playing
by Google

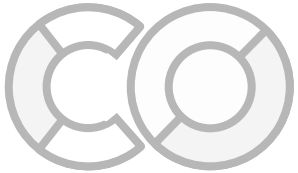


What are we going to learn?



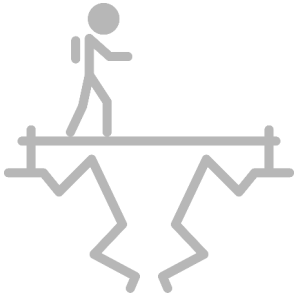
Challenges with Keyword Spotting

The Keyword Spotting ML **Pipeline**

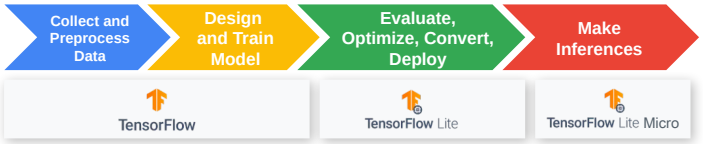


Hands-on training of a Keyword Spotting Model

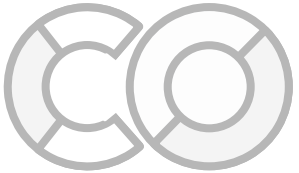
What are we going to learn?



Challenges with a Wake Words Application

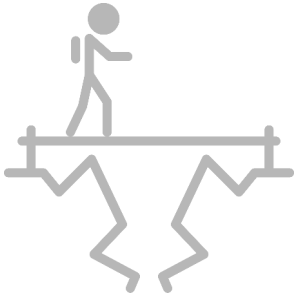


The Keyword Spotting ML **Pipeline**



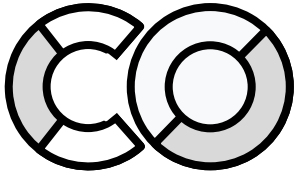
Hands-on training of a wake words model in Colab!

What are we going to learn?



Challenges with a Wake Words Application

The Wake Words ML **Pipeline**



Hands-on training of a Keyword Spotting Model